

Day 3 & Day 4 Assignment

1. Explain in detail about Agile Methodology

Agile Methodology is a software development approach that focuses on continuous improvement, teamwork, customer collaboration, and quick delivery of working software. Instead of developing the entire product at once, Agile divides the project into small iterations called **Sprints**, usually lasting 2–4 weeks. **Agile Values:**

- Individuals and interactions over processes and tools.
- Working software over comprehensive documentation.
- Customer collaboration over contract negotiation.
- Responding to change over following a plan.

Agile Principles:

1. Deliver software frequently.
2. Welcome changing requirements.
3. Business and development teams work together.
4. Build projects around motivated individuals.
5. Maintain a sustainable development pace.
6. Regularly improve through retrospectives.

Agile Life Cycle:

Requirement Gathering → Planning → Sprint Planning → Development → Testing → Review → Release → Retrospective → Next Sprint.

Advantages:

- Faster product delivery.
- Better customer satisfaction.
- Early bug detection.
- High flexibility and adaptability.
- Improved product quality through continuous testing.

Disadvantages:

- Requires active customer involvement.
- Difficult to estimate cost for large projects.
- Needs experienced and collaborative teams.

Conclusion:

Agile is one of the most widely used software development methodologies because it helps teams deliver high-quality software quickly while adapting to changing customer needs.

2. Explain about Monitoring Activity

Monitoring Activity is the process of continuously tracking the progress, quality, schedule, cost, and risks of a software project to ensure that it meets its planned objectives. **Purpose:**

- Track project progress.
- Identify issues early.
- Ensure quality standards are maintained.
- Verify tasks are completed on time.
- Improve team performance.

Monitoring Activities in Agile:

- Daily Scrum meetings.
- Sprint Burndown Chart.
- Sprint Review.
- Sprint Retrospective.

- Defect Tracking and Test Reporting.

Benefits:

- Early detection of risks and defects.
- Better communication among team members.
- Improved project quality.
- On-time delivery.
- Better customer satisfaction.

Conclusion:

Monitoring activity helps ensure that a software project stays on track by continuously evaluating progress, quality, and performance, enabling teams to take corrective actions whenever required.